PAVIT SINGH NAGPAL

Mail: nagpalpavitsingh@gmail.com **Phone**: +1(303)214-8958
Linkedin: https://www.linkedin.com/in/pavit-singh-nagpal-67287611a/ **Github**: https://github.com/pavit2000

TECHNICAL PROFICIENCY:

Languages: JavaScript, Java, Python, C#, C, C++, TypeScript **Databases:** PostgreSQL, MSSQL, MongoDB
Tools and Technology: React, Node.js, AWS, Hadoop, Spark, Docker, Kubernetes, REST API, GitHub, Linux, Jira, Unity

PROFESSIONAL EXPERIENCE:

Holiday Channel – Software Engineer **Jul 2024 – Present**

- Led a team of 5 to develop a budget tracker tool that helps users plan and manage travel expenses efficiently. The tool features real-time budget tracking, spending insights, and collaborative sharing options, leading to a 10% increase in user base.
- Developed and optimized responsive front-end components using React, TypeScript, Tailwind CSS, and shadcn, collaborating with UI design and back-end teams to ensure a seamless user experience, resulting in a 1.15x increase in user engagement.
- Implemented interactive data visualizations using Recharts, allowing users to gain deeper insights into their budget and spending patterns.
- Designed RESTful and GraphQL APIs using Node.js, ensuring secure, high-performance request handling and data flow.
- Utilized Supabase for data storage, leveraging its real-time capabilities and PostgreSQL foundation to ensure efficient, scalable, and secure data management.
- Integrated Okta services for Single Sign-On (SSO) to ensure secure user login data and enable seamless authentication across tools like the budget tracker.

Medtronic – Capstone Project (Transcatheter Mitral Valve Placement Training Simulator) **Oct 2023 – Apr 2024**

- Developed a simulator for valve placement training for cardiologists, enhancing procedural training and reducing reliance on physical models.
- Implemented an optimized catheter pathfinding algorithm using Unity's Splines package, improving placement accuracy by 25% in the simulation.
- Created C# scripts to control catheter deflection, rotation, and movement, ensuring realistic interaction and precise maneuverability.
- Built high-fidelity 3D models using Unity and Blender, focusing on user interaction and immersive visualization.
- Collaborated with Medtronic engineers and clinicians to tackle key challenges in valve replacement training, ensuring clinical accuracy and usability.

ACADEMIC PROJECTS:

Kubernetes-Powered Music Separation Service with Min.io Integration **Dec 2023**

- Engineered a Kubernetes-powered music separation service with Docker-based worker nodes, automating audio component isolation and reducing processing time by 40%.
- Optimized task distribution using Redis queues and integrated Min.io for efficient data handling, enabling scalability to process 100+ audio files concurrently with minimal latency in a Docker and Kubernetes environment.

Fitness Tracker Web Application **Apr 2023**

- Built a fitness tracker web application using Python, Flask, and MongoDB, implementing a data-driven algorithm to analyze heart rate, step counts, and workout trends, improving activity tracking accuracy by 35%.
- Integrated RabbitMQ for asynchronous messaging and SMTP for automated email reminders, reducing notification delivery time by 50%, while leveraging GitHub Actions and Heroku for seamless CI/CD deployment.

A Spatiotemporal Context Aware Hierarchical Model for Corporate Bankruptcy Prediction **Jul 2021**

- Developed a spatiotemporal context-aware hierarchical model for corporate bankruptcy prediction, achieving 97% accuracy and an F1 score of 0.95, which improved early bankruptcy detection by 40%, leveraging Knowledge Graph embeddings, Machine Learning algorithms, and Bidirectional Attention-based LSTM.
- Designed an algorithm to detect financial distress by analyzing 25,000 news articles and 20+ financial metrics, enabling a comprehensive risk assessment framework that reduced false positives by 30%, enhancing financial decision-making accuracy.

EDUCATION:

Masters in Computer Science: University of Colorado Boulder GPA: **3.77/4** **Aug 2022 – May 2024**
Coursework: Design and Analysis of Algorithms, Natural Language Processing (NLP), Numerical Computation, Linear Programming, Foundations of Software Engineering, Data Mining, Datacenter Scale Computing, Leading Oneself

B. Tech. in Computer Science: Kalinga Institute of Industrial Technology (KIIT) GPA: **9.06/10** **Jul 2018 – May 2022**